

Phi

A constructed language

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# 1 Introduction

Phi is a constructed language. A constructed language is one whose phonology, grammar, and vocabulary, instead of having developed naturally, are consciously devised<sup>1</sup>. The sound of the name *Phi* embodies the simplicity and softness of the language itself. *Phi* ( $\Phi$ ) is also beauty in mathematics. It is

*“the golden ratio, an irrational number with a value of approximately 1.618033988 which expresses the relationship that the sum of two quantities is to the larger quantity as the larger is to the smaller.” – Wiktionary<sup>2</sup>*

It is thought that the value of *phi* is one of the fundamental characteristics of nature; in the language Phi, nature is fundamental to its being. Listening to spoken Phi one is sometimes reminded of the sounds of the natural world. There is also a musicality and rhythm to sentences in the language.

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1 [https://en.wikipedia.org/wiki/Constructed\\_language](https://en.wikipedia.org/wiki/Constructed_language)

2 <https://en.wiktionary.org/wiki/%CF%86>

## 2 The Tengwar

Phi can be written in either of two ways: using the standard Latin alphabet or the Tengwar script. Tengwar was invented by J. R. R. Tolkien and first described in Appendix E of the third book in his trilogy, *The Lord of the Rings*. There are numerous sources of information available concerning the Tengwar so details and history of the script will not be explored here.

The look and feel of the Tengwar characters (*tengwa*) and its diacritical marks (*tehta*) match the ideals and philosophies behind this language. Phi has a small set of consonants and vowels so it is relatively straightforward to map its graphemes to Tengwar characters. The Unicode font *Tengwar Telcontar*<sup>3</sup> is used in this manual to render Phi text in Tengwar script. A custom keyboard layout<sup>4</sup> was employed to facilitate typing the Tengwar characters.

Tables 1 and 2 outline the mapping between the graphemes used in Phi and the corresponding Tengwar characters, as well as the keypresses necessary to produce the characters using the font and keyboard layout mentioned previously. An application or utility that supports Graphite fonts<sup>5</sup> is required for the font to render correctly. This manual was produced using LibreOffice Writer v7.1.2.2, which renders *Tengwar Telcontar* properly. A link to details of this requirement can be found in the first footnote below.

Written Phi uses the Tengwar's Quenya mode. Vowel diacriticals are placed on the consonant that precedes the vowel. Since all Phi words start with a consonant, there is never a need to use an initial vowel carrier. Words in Phi often contain two consecutive vowels, but a vowel carrier is still unnecessary: the second vowel is placed below the same consonant. The letters would then be read as consonant, upper vowel, lower vowel. This and the use of a single Tengwar character for consonant doubles such as *ph* or *sh* cause Phi script to be very compact and efficient.

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3 <http://freetengwar.sourceforge.net/tengtelc.html>

4 <http://freetengwar.sourceforge.net/keylayouts.html>

5 [https://scripts.sil.org/cms/scripts/page.php?site\\_id=projects&item\\_id=graphite\\_fonts](https://scripts.sil.org/cms/scripts/page.php?site_id=projects&item_id=graphite_fonts)

The Phi word for “love” is *lothea*. In the Tengwar script it looks like this:



A six-letter word in latin characters requires only two *tengwa* to represent it. Many words in Phi are two to three characters long. This occurs mainly with parts of speech such as particles, prepositions, or pronouns. These words are almost always a single *tengwa* with diacriticals in the Tengwar. All of this results in a very compact but clear Phi writing system.

This is the phrase “I love you” in its simplest vernacular form in Phi:



That is a beautiful sentiment rendered in a beautiful script. In more formal communication, particles are used to mark parts of speech in order to reduce ambiguity. As well, word markers are often used to distinguish multisyllabic words for clarity. Phi Tengwar punctuation is outlined in Table 3. As an example, the phrase “I love you” would be written formally as:

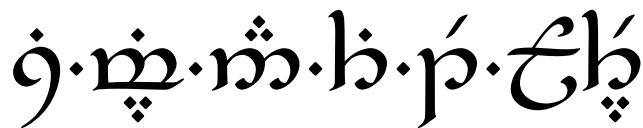


Table 1: Phi consonant grapheme to Tengwar mapping

| Grapheme | Tengwa | Keypress |
|----------|--------|----------|
| p        | ᑭ      | p        |
| t        | ᑭ      | t        |
| m        | ᑭ      | m        |
| n        | ᑭ      | n        |
| r        | ᑭ      | shift-r  |
| s        | ᑭ      | s        |
| s        | ᑭ      | shift-s  |
| h        | ᑭ      | h        |
| l        | ᑭ      | l        |
| w        | ᑭ      | w        |
| ph       | ᑭ      | shift-p  |
| th       | ᑭ      | shift-t  |
| sh       | ᑭ      | shift-c  |
| wh       | ᑭ      | shift-k  |

Table 2: Phi vowel grapheme to Tengwar mapping

| Grapheme | Tehta                                                                             | Keypress                  |
|----------|-----------------------------------------------------------------------------------|---------------------------|
| i        |  | i / option-i <sup>6</sup> |
| e        |  | e / option-e              |
| o        |  | o / option-o              |
| a        |  | a / option-a              |

Table 3: Phi punctuation to Tengwar mapping

| Punctuation | Tengwar | Keypress |
|-------------|---------|----------|
| space       | •       | ,        |
| period      | ⋮       | shift-,  |

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<sup>6</sup> Keypresses for the low diacriticals are given for macOS. Use ctrl-alt on Windows instead of option.

### 3 Examples

Following are several examples of Phi phrases in formal mode. They are given first in English, then in romanized text, and finally in the Tengwar script.

My name is Daniel

*si phiathe soa mia na thanie te phi*

ጳጳሱ ጳጳሱ ጳጳሱ ጳጳሱ ጳጳሱ ጳጳሱ ጳጳሱ

I love you.

*si mia na thi te lothea*

፳፻፲፱.፳፻፲፱.፳፻፲፱.፳፻፲፱.፳፻፲፱.፳፻፲፱

The sky is blue.

*si whetho na mipho te phi*

ገ.ብ.ጽ.ጠ.ጥ.ገ

They like food.

*si lo sha na noshea te lephe*

၎င်းတို့သည် နေရာအနှံ့တွင် နေထိုင်ကြသည်။

I like it.

*si mia na sha te lephe*

᠑ᠡᠮᠡᠨᠠᠵᠤᠰᠤᠨᠵᠢᠨ

I do no like green!

*ho si mia na hashe te me lephe*

ጸ.፻.፱.፳.፻፲.፲፩.፲፪.፲፫

I liked my friend.

*si mia na miwho soa mia te li lephe*

፩.፵.፳.፵.፩.፪.፶.፶.፶.፶

The food is ours.

*si noshea na soa lo mia te phi*

፩.፳፩.፳.፪.፶.፶.፶.፶

Our food is good.

*si noshea soa lo mia na shishoa te phi*

፩.፳፩.፪.፶.፶.፶.፳፩.፶.፶

Are you in good health?

*si wha thi na ropho shishoa te la phe*

፩.፩.፳.፳.፳፩.፳፩.፶.፶.፶

We love the orange trees.

*si lo mia na lo lipha sathe te lothea*

፩.፶.፶.፳.፶.፶.፩.፶.፶.፶